

Radio-Frequency-Interference Rejection for Pulsed NQR

Reference-coil cancellation, vector pickup, and gated acquisition

Abstract

Ambient radio-frequency interference (RFI) is a practical limiting factor for low-frequency ^{14}N nuclear-quadrupole-resonance (NQR) experiments. The difficult case is coherent interference, often from AM broadcast stations or nearby electronics, that overlaps the NQR detection band. This note summarizes candidate mitigation methods and proposes a simulation roadmap for MRSpinDynamics. The central recommendation is to model RFI as a vector magnetic field and to support multi-channel reference-coil cancellation from the beginning. In pulsed NQR sequences such as SLSE and SORC, the NQR receiver is only scientifically useful in the gaps between RF pulses, whereas reference detectors may run continuously; the implementation should therefore treat cancellation as a masked/gated estimation problem rather than as ordinary continuous denoising of a single sampled waveform.

1 Why this belongs in the NQR model

^{14}N NQR lines for energetic materials and other nitrogenous solids can lie in the same frequency range as strong environmental emitters. In the RAND landmine-detection review, the TNT NQR frequencies are reported as roughly 790–900 kHz, directly in the AM broadcast band; the review states that environmental RF interference can overwhelm TNT responses and that practical NQR systems have used coil design plus a separate RFI monitoring antenna followed by subtraction [1]. A later appendix in the same review calls out commercial AM stations within about 20 kHz of the TNT detection frequencies and nearby electromagnetic noise sources as current NQR limitations [1].

This is not only a filtering problem. The desired NQR signal is transient, weak, sequence-gated, and often detected with matched filters or echo integration. A stationary notch filter may reduce a carrier but can also distort the baseline, remove useful spectral content, or leave time-varying AM sidebands. A physically faithful simulator should instead expose the experimenter to the trade space among passive rejection, reference-channel subtraction, active cancellation, pulse gating, and detection bias.

2 Signal model

Let the primary NQR receive channel after analog front-end conditioning be

$$y[n] = s_{\text{NQR}}[n] m[n] + \sum_{q=0}^{Q-1} (g_q * u_q)[n] + \eta_y[n], \quad (1)$$

where s_{NQR} is the desired NQR response, $m[n] \in \{0, 1\}$ is the receive-valid mask that removes RF-pulse and ringdown intervals, u_q are external interference sources, g_q are their transfer functions into the primary channel, and η_y is receiver noise.

For a reference-coil system, the measured reference vector is

$$\mathbf{x}[n] = \begin{bmatrix} x_1[n] & x_2[n] & \cdots & x_K[n] \end{bmatrix}^T, \quad (2)$$

and the cleaned primary channel is estimated as

$$\hat{s}_{\text{NQR}}[n] = y[n] - \sum_{k=1}^K (h_k * x_k)[n]. \quad (3)$$

The filters h_k may be complex scalars, short FIR filters, frequency-domain transfer functions, or adaptive filters. This is the NQR analogue of adaptive noise cancellation [2, 3], RFI reference-antenna cancellation in radio astronomy [4, 5], and external-sensor EMI subtraction in low-field MRI [6].

3 The magnetic field is a vector

A single reference coil measures only one projection of the RFI magnetic field. For a small loop with effective pickup vector $\mathbf{c}_k$, the idealized reference voltage is proportional to

$$x_k(t) \propto \frac{d}{dt} \int_{A_k} \mathbf{B}_{\text{RFI}}(\mathbf{r}, t) \cdot d\mathbf{A} \approx \mathbf{c}_k^T \frac{d\mathbf{B}_{\text{RFI}}(\mathbf{r}_k, t)}{dt}. \quad (4)$$

The primary NQR receive coil has its own pickup vector $\mathbf{c}_y$. In the far-field or spatially uniform limit,

$$y_{\text{RFI}}(t) \approx \mathbf{c}_y^T \dot{\mathbf{B}}_{\text{RFI}}(t), \quad \mathbf{x}(t) \approx C^T \dot{\mathbf{B}}_{\text{RFI}}(t), \quad (5)$$

where $C = [\mathbf{c}_1, \dots, \mathbf{c}_K]$. If C spans the component of $\dot{\mathbf{B}}_{\text{RFI}}$ that couples into the primary coil, the primary interference projection can be estimated. If not, the cancellation is rank-deficient.

For distant AM transmitters, the wavelength is hundreds of meters and the field may be nearly uniform over the probe. Three approximately orthogonal reference coils near the instrument can then be a reasonable first-order model. For local electronics, shield currents, cable pickup, or re-radiating metal, the RFI can be near-field and spatially structured. A single three-axis reference station may then fail; the next model is multiple vector stations, with $K = 3P$ channels for P physical reference locations. Electric-field pickup can also enter through cabling and preamplifier imperfections, so the software model should allow reference channels that are not purely magnetic.

4 Pulsed acquisition changes the estimation problem

In SLSE, SORC, and related pulsed NQR sequences, the primary receive channel is blanked or unusable during transmit pulses and during subsequent ringdown. The desired NQR signal is only acquired in gaps. Reference detectors, by contrast, can often run continuously because they are physically decoupled from the NQR transmit/receive coil or can tolerate the pulse feedthrough.

The cancellation problem should therefore be written with masks. Let $\mathcal{G}$ be the set of valid receive samples in the pulse gaps and $\mathcal{B}$ be a set of samples believed to contain no NQR response, such as pre-shot baselines, late-time tails, off-resonance shots, or inter-scan calibration windows. A fixed FIR canceller can be estimated by

$$\hat{\mathbf{h}} = \arg \min_{\mathbf{h}} \sum_{n \in \mathcal{B}} |y[n] - \Phi_x[n]\mathbf{h}|^2 + \lambda \|\mathbf{h}\|_2^2, \quad (6)$$

where $\Phi_x[n]$ is the tapped-delay design row assembled from the continuous reference channels. The cleaned NQR data are then computed only on $\mathcal{G}$. Continuous reference acquisition is still useful because it allows:

- interpolation of the RFI phase and amplitude across transmit blanking;
- tracking of AM modulation envelopes and beat notes over the full shot;
- estimation from pulse-contaminated reference intervals while excluding those intervals from the primary objective;
- transfer-function updates between shots without corrupting the NQR response in the detection gaps.

This masking is essential for simulation. A method that appears to work on continuous synthetic FIDs may fail once the primary channel only contributes short windows separated by high-power RF pulses.

5 Candidate RFI-rejection methods

6 Active cancellation architecture

Active RFI cancellation adds one or more drive coils with command vector $\mathbf{a}[n]$. The primary channel becomes

$$y[n] = s_{\text{NQR}}[n]m[n] + y_{\text{RFI}}[n] + (G_a * \mathbf{a})[n] + \eta_y[n]. \quad (7)$$

The controller chooses $\mathbf{a}$ to minimize the primary RFI estimate, using reference coils and calibration of the actuator-to-sensor transfer functions. The natural simulation layers are:

1. feedforward cancellation, where references predict the required actuator waveform without using the primary NQR channel as an error signal during receive gaps;
2. feedback cancellation during baseline windows, with adaptation frozen during expected NQR signal windows;

Table 1: RFI mitigation methods to represent in MRSpinDynamics.

Method	Useful model	Main failure mode
Passive shielding and cabling hygiene	Lower the coupled g_q before digitization; low-field MRI work emphasizes diagnostic noise measurements, grounding, cabling, and shielding as prerequisites for reaching coil-noise limits [7].	Does not remove unavoidable broadcast pickup; cannot be studied as a digital post-processing algorithm alone.
Coil balancing or gradiometry	Common-mode rejection of spatially smooth ambient magnetic fields, at the cost of changing sample sensitivity.	Near-field RFI gradients and sample-signal cancellation.
Static notch or spectral excision	Remove known narrowband carriers after acquisition.	Signal distortion, AM sidebands, ringing, and data loss near the NQR line.
Reference-channel least squares or Wiener cancellation	Estimate Eq. (3) from baseline windows, off-resonance shots, or late-time gaps.	Reference channels do not span the primary RFI; reference noise is injected.
Adaptive LMS/NLMS/RLS cancellation	Track slowly changing interference amplitude, phase, and transfer functions. This is the classical adaptive-noise-canceller family [2, 3].	Adaptation can subtract NQR signal if trained on contaminated windows; LMS may converge too slowly or misadjust under low reference SNR.
Multi-reference cancellation	Use several reference channels or vector stations; radio astronomy analyses show that two-reference structures avoid some residual-RFI floors that remain in single-reference cancellers [4].	More reference noise and more calibration degrees of freedom.
AM carrier/envelope tracking	Represent AM stations as carrier plus sidebands/envelope and track a compact parametric model.	Breaks when interference is broadband, nonstationary, or multi-source.
Active cancellation coils	Drive compensation coils to null the field at the receive coil, optionally with digital post-cleanup.	Loop stability, actuator saturation, transmit-pulse feedthrough, and accidental nulling of useful sample signal.

3. hybrid analog/digital cancellation, where active coils prevent front-end saturation and digital reference subtraction removes residuals.

The simulator should explicitly represent actuator limits and closed-loop stability. Digital post-processing can repair additive contamination after linear acquisition, but it cannot recover data lost to LNA compression, ADC clipping, or transmit-pulse recovery artifacts.

7 Simulation observables

The RFI module should report more than a cleaned waveform. Useful diagnostics include:

- RFI suppression in dB, measured in the primary receive gaps;
- matched-filter or echo-integral NQR SNR before and after cancellation;
- NQR amplitude and phase bias caused by cancellation;
- residual spectral lines and residual autocorrelation;
- reference-noise injection into the cleaned primary channel;
- conditioning/rank of the reference-channel design matrix;
- fraction of samples masked by transmit pulses and ringdown;
- clipping or front-end saturation flags before digital cancellation;
- false-alarm and detection-probability changes for a chosen detector.

8 Implementation plan for MRSpinDynamics

8.1 Phase 1: synthetic RFI and reference channels

Add a small signal-processing layer that can generate coherent RFI sources and reference measurements independent of any NQR sequence:

- narrowband tones, AM-modulated carriers, chirps, broadband colored noise, and impulsive local interference;
- vector magnetic-field sources $\mathbf{B}_{\text{RFI}}(t)$, with optional electric pickup channels;
- reference coil objects with pickup vectors, locations, bandwidths, noise floors, gain/phase errors, saturation limits, and optional NQR leakage;
- primary receive pickup vectors coupled to existing NQR orientation and receiver conventions.

8.2 Phase 2: masks for pulsed NQR sequences

Expose receive-valid masks from SLSE, SORC, FID, and full-density-matrix NQR simulations. The mask should distinguish transmit pulse, ringdown, valid receive, and optional baseline-only windows. Reference channels should be sampled continuously over the same shot clock. This allows a cancellation algorithm to train on $\mathcal{B}$, apply on $\mathcal{G}$, and ignore samples where the primary receiver would be blanked.

8.3 Phase 3: cancellers

Implement cancellers as pure numerical transforms with explicit fit/apply separation:

- scalar complex subtraction for a single reference channel;
- multi-channel ridge least-squares FIR cancellation using Eq. (6);
- frequency-domain transfer-function cancellation for stationary RFI;
- LMS/NLMS and RLS adaptive cancellers with mask-aware freeze/update controls;
- parametric AM carrier trackers as a baseline for broadcast-like RFI.

Each canceller should return coefficients, cleaned data, predicted RFI, reference-noise estimates, and mask-aware diagnostics.

8.4 Phase 4: active cancellation

Add optional compensation coils after the passive/digital path is validated. The first implementation can be feedforward with known actuator transfer functions. Later versions can add feedback adaptation, stability margins, and closed-loop interaction between compensation coils, reference sensors, and the primary receive coil.

8.5 Phase 5: examples and tests

Examples should start with intentionally simple cases and then add the pathologies that matter experimentally:

1. a single AM carrier near a ^{14}N line with one reference coil;
2. three-axis reference coils cancelling a time-varying RFI polarization;
3. rank-deficient one-axis cancellation that leaves a residual in the primary coil;
4. SLSE receive gaps with continuous references and transmit/ringdown masks;
5. reference-channel NQR leakage showing amplitude bias;
6. preamplifier or ADC clipping showing when digital cancellation is too late;
7. active feedforward cancellation followed by digital residual cleanup.

Unit tests should verify analytic least-squares recovery in noise-free cases, known dB suppression in seeded stochastic cases, no update during masked NQR windows, and preservation of a synthetic NQR matched-filter amplitude when reference leakage is zero.

9 Open design questions

- Should RFI live under `spin_dynamics.nqr` or in a shared `spin_dynamics.interference` namespace usable by low-field MRI and ESR examples?
- Should reference coils use the existing orientation objects, or should they have their own lab-frame geometry model?

- What is the right default for transmit-pulse feedthrough into reference channels: blank, clip, saturate, or preserve for controller tracking?
- How should matched-filter detection report uncertainty after adaptive cancellation, given injected reference noise and fitted degrees of freedom?
- How much active-cancellation control theory belongs in the package versus being represented as calibrated transfer functions and actuator limits?

References

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